

04_REFERENCE.md

SpeechStudio Higgs Audio V3 Reference

Version: 1.0

Status: Approved

1. Purpose

This document serves as the complete reference for every Higgs Audio V3 capability supported by SpeechStudio.

It defines every officially supported control token, generation parameter and verified behaviour used by the application.

This document is the primary reference for both developers and AI coding assistants.

2. Supported Model

Current backend

multimodalart/higgs-audio-v3-tts-4b-transformers

Original model

BosonAI Higgs Audio V3 TTS 4B

Execution

Transformers

CUDA

3. Emotion Tokens

SpeechStudio supports every official Higgs emotion.

Emotion	Token	Description
Elation	<\ emotion:elation\ >	Joy

Emotion	Token	Description
Amusement	<code><\ emotion:amusement\ ></code>	Playful
Enthusiasm	<code><\ emotion:enthusiasm\ ></code>	Excitement
Determination	<code><\ emotion:determination\ ></code>	Firm
Pride	<code><\ emotion:pride\ ></code>	Confident
Contentment	<code><\ emotion:contentment\ ></code>	Calm satisfaction
Affection	<code><\ emotion:affection\ ></code>	Warm
Relief	<code><\ emotion:relief\ ></code>	Relief
Contemplation	<code><\ emotion:contemplation\ ></code>	Reflective
Confusion	<code><\ emotion:confusion\ ></code>	Confused
Surprise	<code><\ emotion:surprise\ ></code>	Surprise
Awe	<code><\ emotion:awe\ ></code>	Wonder
Longing	<code><\ emotion:longing\ ></code>	Yearning
Arousal	<code><\ emotion:arousal\ ></code>	Heightened intensity
Anger	<code><\ emotion:anger\ ></code>	Anger
Fear	<code><\ emotion:fear\ ></code>	Fear
Disgust	<code><\ emotion:disgust\ ></code>	Disgust
Bitterness	<code><\ emotion:bitterness\ ></code>	Bitterness
Sadness	<code><\ emotion:sadness\ ></code>	Sadness
Shame	<code><\ emotion:shame\ ></code>	Shame
Helplessness	<code><\ emotion:helplessness\ ></code>	Helpless

4. Style Tokens

Style	Token
Singing	<code><\ style:singing\ ></code>
Whispering	<code><\ style:whispering\ ></code>
Shouting	<code><\ style:shouting\ ></code>

5. Prosody Tokens

Speed

Token	Effect
speed_very_slow	~0.65×
speed_slow	~0.85×
speed_fast	~1.2×
speed_very_fast	~1.4×

Pitch

Token	Effect
pitch_low	Lower pitch
pitch_high	Higher pitch

Delivery

Token	Effect
expressive_high	More expressive
expressive_low	Flatter delivery

Pauses

Token	Effect
pause	Short pause
long_pause	Long pause

6. Sound Effects

Effect	Token	Suggested Text
Cough	<\ sfx:cough\ >	Ahem
Laughter	<\ sfx:laughter\ >	Haha
Crying	<\ sfx:crying\ >	Sob
Screaming	<\ sfx:screaming\ >	Ahh

Effect	Token	Suggested Text
Burping	<code><\ sfx:burping\ ></code>	Burp
Humming	<code><\ sfx:humming\ ></code>	Hmm
Sigh	<code><\ sfx:sigh\ ></code>	Ahh
Sniff	<code><\ sfx:sniff\ ></code>	Sff
Sneeze	<code><\ sfx:sneeze\ ></code>	Achoo

7. Generation Parameters

Parameter	Type	Default	UI Range
temperature	float	1.0	0.1–2.0
top_p	float	0.95	0.1–1.0
top_k	integer	50	1–500
max_new_tokens	integer	2048	128–4096
seed	integer	Random	Any
append_silence	float	1.0 s	0–5 s

8. Voice Cloning

Supported

Reference WAV

Reference Transcript

Reference Sample Rate

Reference Audio Validation

Voice Profiles

9. Prompt Rules

Sentence level tokens

Emotion

Style

Pitch

Speed

Delivery

Inline tokens

Pause

Long Pause

Sound Effects

10. Verified SpeechStudio Findings

This section contains experimentally verified behaviour.

Examples

Approximately 15 seconds of speech is consistently reliable.

Appending one second of silence improves natural sentence endings.

Emotion transitions produce more stable output than combining multiple emotions within a single sentence.

Voice cloning quality improves with clean reference recordings.

11. GUI Mapping

Every supported Higgs feature is mapped to a dedicated UI control.

Emotion

Emotion Panel

Style

Style Panel

Prosody

Prosody Panel

Sound Effects

SFX Panel

Generation Parameters

Generation Panel

Reference Voice

Voice Panel

Prompt Preview

Prompt Preview Panel

12. Prompt Examples

Short narrator

Podcast

Movie trailer

Conversation

Horror

Documentary

Fast speech

Whisper

Singing

Voice clone

These examples are maintained as living examples and updated when verified improvements are found.

13. Official References

BosonAI Higgs Audio

PROMPTING.md

Hugging Face Model Card

Transformers Port

SpeechStudio Deep Research

SpeechStudio Verified Experiments